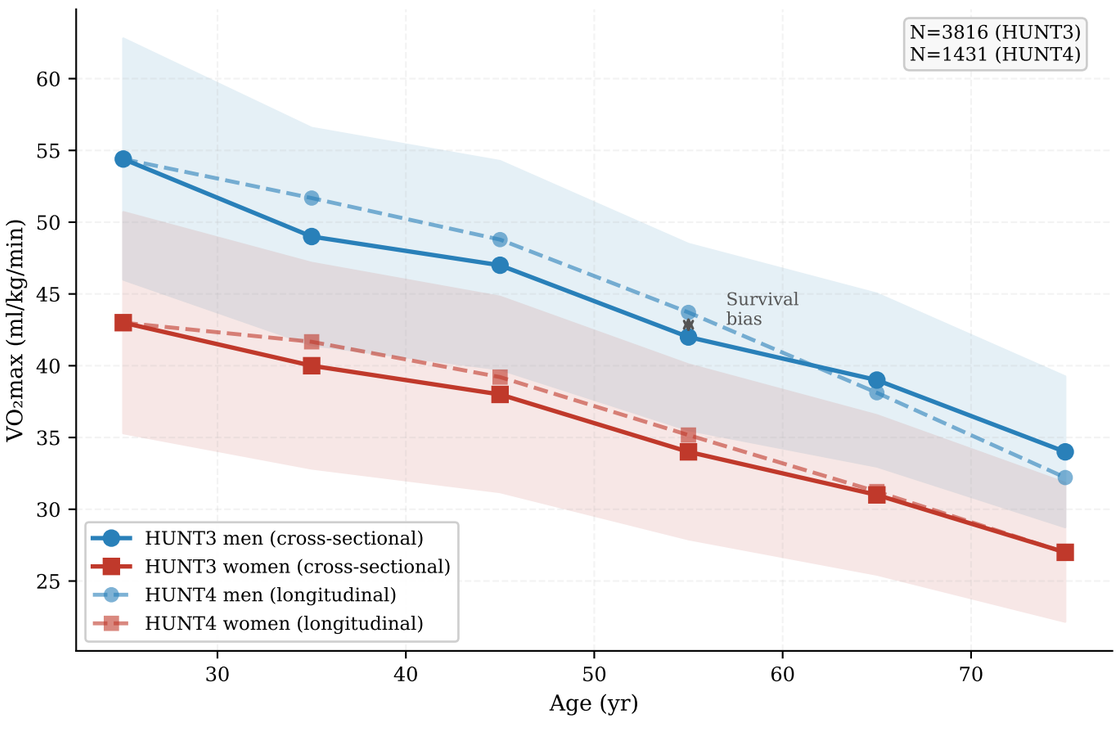
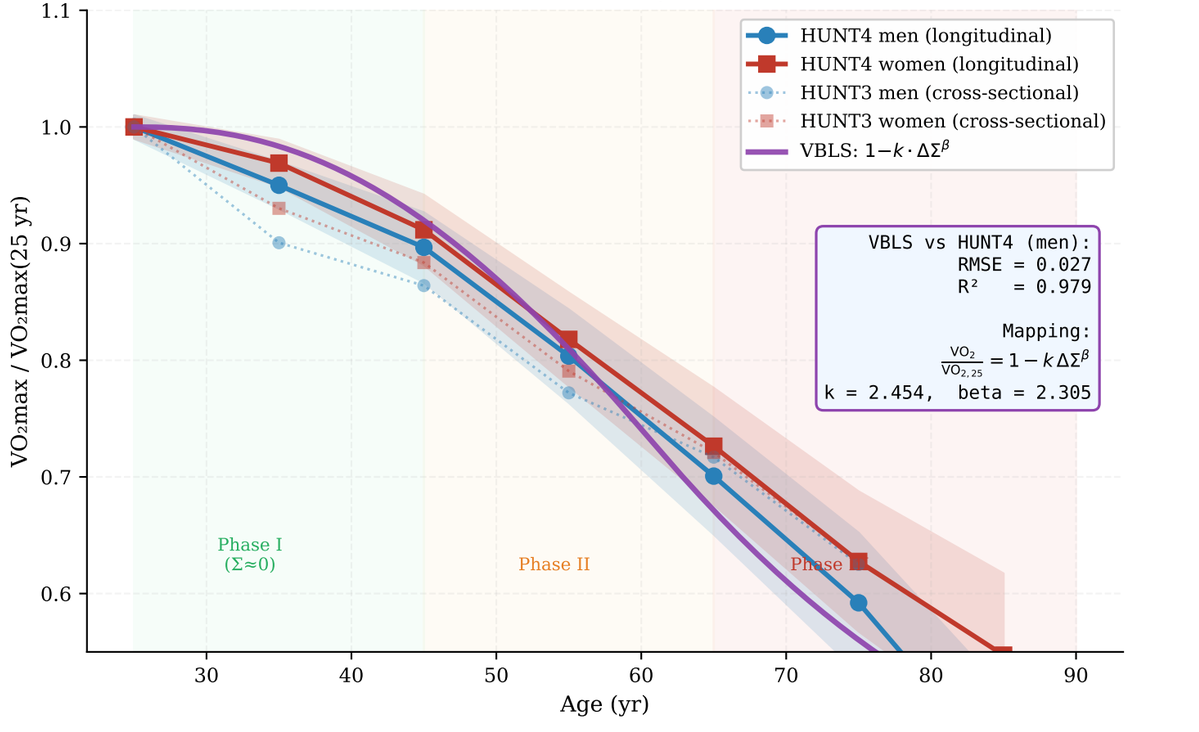
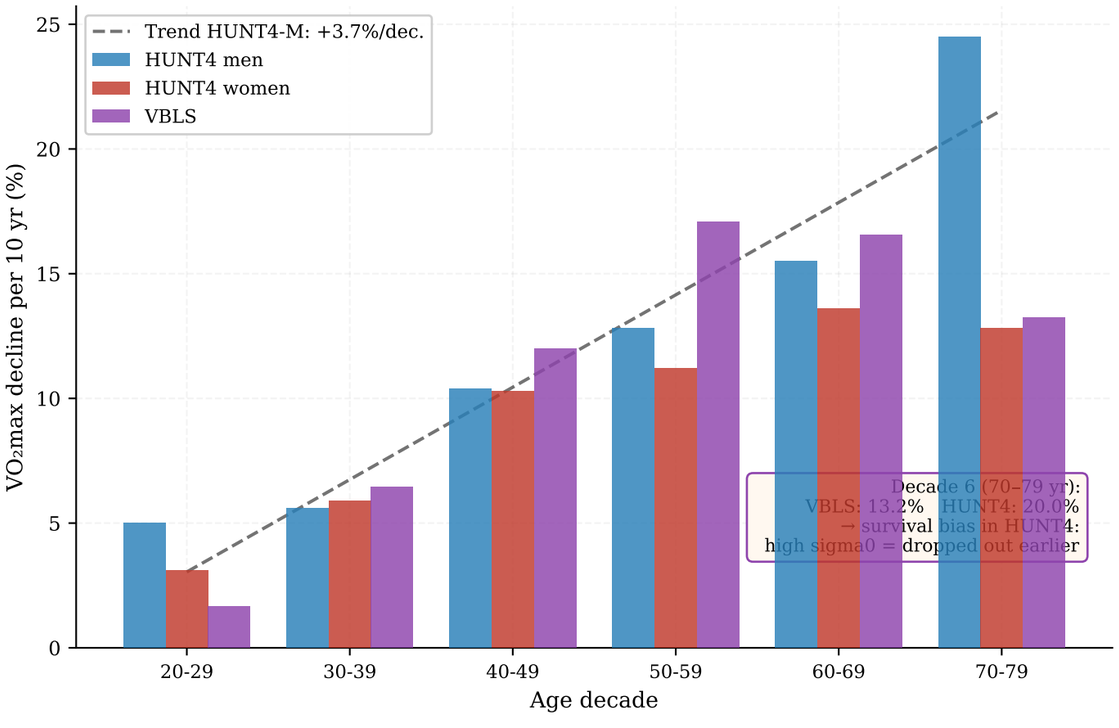
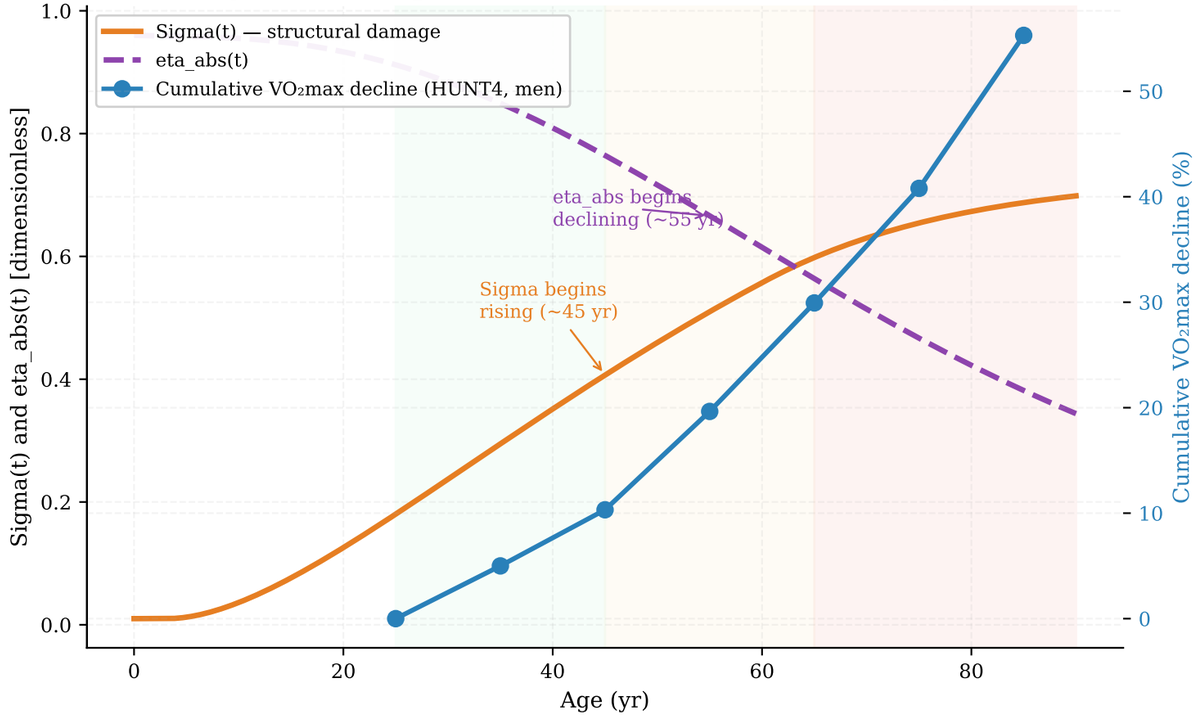

A. VO₂max by age: HUNT3 vs HUNT4
Absolute values
N=3816 (HUNT3)
N=1431 (HUNT4)
Survival bias
HUNT3 men (cross-sectional)
HUNT3 women (cross-sectional)
HUNT4 men (longitudinal)
HUNT4 women (longitudinal)
VO₂max (ml/kg/min)
Age (yr)


B. Normalized VO₂max decline
VBLS vs HUNT3/HUNT4
HUNT4 men (longitudinal)
HUNT4 women (longitudinal)
HUNT3 men (cross-sectional)
HUNT3 women (cross-sectional)
VBLS: 1−k·ΔΣ^β
VBLS vs HUNT4 (men):
RMSE = 0.027
R² = 0.979
Mapping:
VO₂ / VO₂,₂₅ = 1 − kΔΣ^β
k = 2.454, beta = 2.305
Phase I
(Σ≈0)
Phase II
Phase
VO₂max / VO₂max(25 yr)
Age (yr)


C. Decade-wise acceleration of VO₂max decline
HUNT4 + VBLS
Trend HUNT4-M: +3.7%/dec.
HUNT4 men
HUNT4 women
VBLS
Decade 6 (70–79 yr):
VBLS: 13.2% HUNT4: 20.0%
→ survival bias in HUNT4:
high sigma0 = dropped out earlier
VO₂max decline per 10 yr (%)
Age decade


D. VBLS dynamics and VO₂max decline
Delay between "Sigma rises" and "eta_abs falls" = memory effect
Sigma(t) — structural damage
eta_abs(t)
Cumulative VO₂max decline (HUNT4, men)
eta_abs begins
declining (~55 yr)
Sigma begins
rising (~45 yr)
Sigma(t) and eta_abs(t) [dimensionless]
Cumulative VO₂max decline (%)
Age (yr)